Class 6 Science – Chapter 12: Beyond Earth

Simple English Answers • Important Questions and Answers • Exam Revision

इस अध्याय के PDF में दिए गए विषयों, चित्रों, तालिकाओं, Think and Answer तथा Quick Check के आधार पर नीचे सरल अंग्रेजी में उत्तर दिए गए हैं। हर मुख्य टॉपिक से अतिरिक्त महत्वपूर्ण प्रश्न भी शामिल हैं।

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1. Stargazing – Questions and Answers

Q1. What might Yukti have seen in the sky?

Answer: Yukti might have seen stars, constellations, planets and other celestial bodies.

Q2. Will binoculars help Yukti?

Answer: Yes, binoculars can help her see some distant objects in the sky more clearly.

Q3. Why are stars more visible in villages than in cities?

Answer: Villages usually have less artificial light and air pollution than cities. Therefore, stars are easier to see in villages.

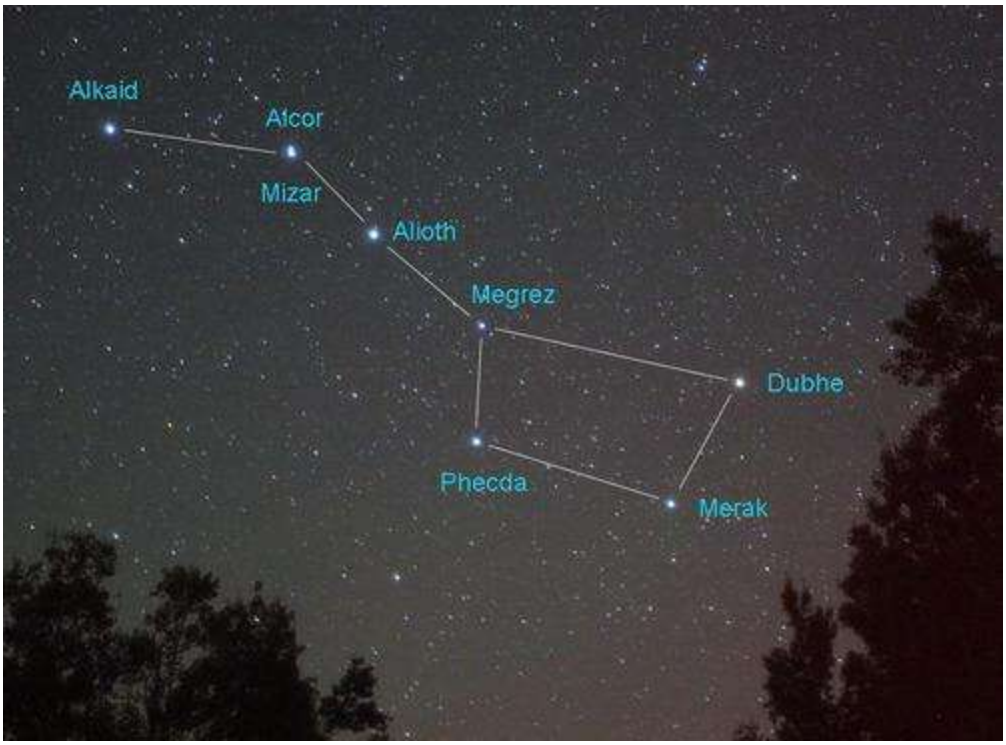
Q4. Which star is the brightest in the night sky?

Answer: Sirius is the brightest star in the night sky.

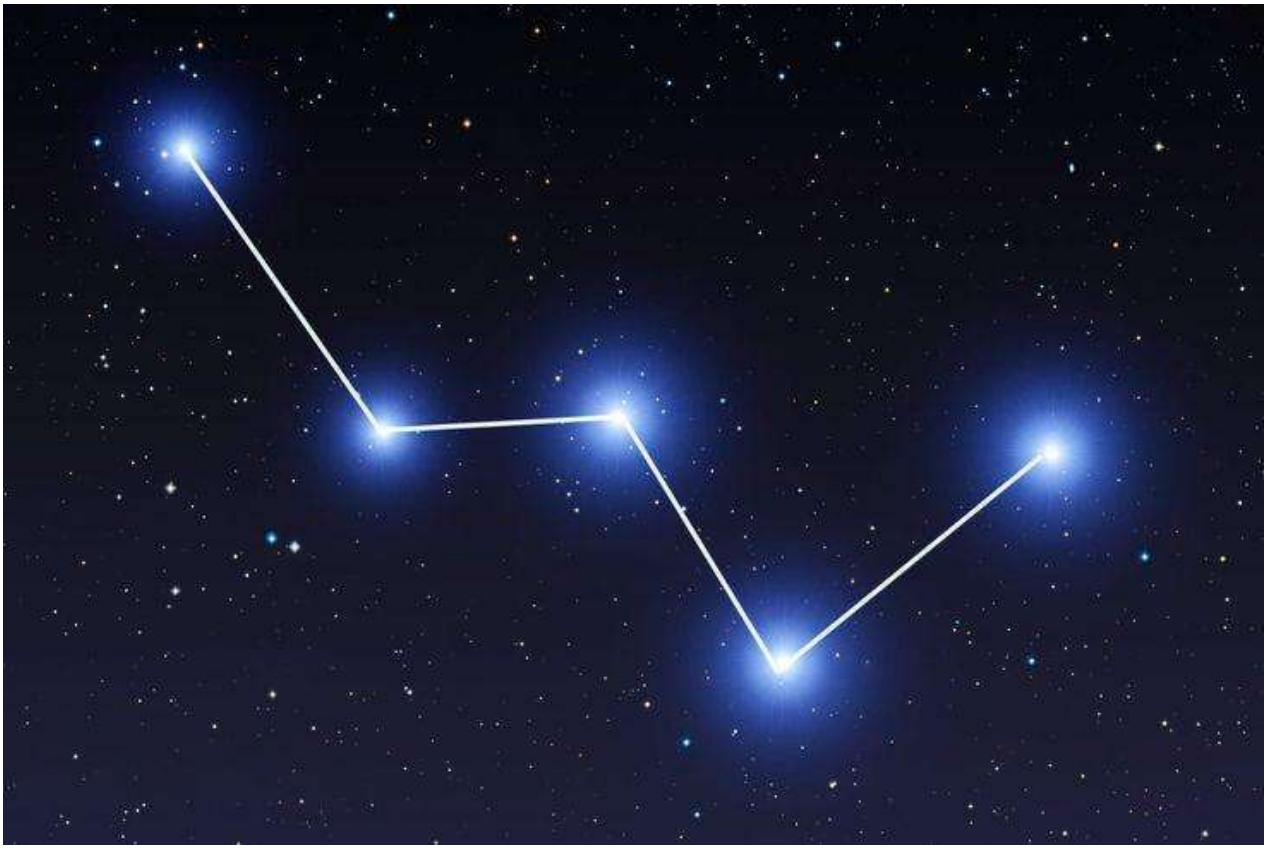
Q5. What things in the sky may not be visible to the naked eye?

Answer: Distant stars, faint galaxies, some planets and other faint celestial objects may not be visible to the naked eye.

2. Stars and Constellations


Alkaid
Alcor
Mizar
Alioth
Megrez
Dubhe
Phecda
Merak





Q1. What is a star?

Answer: A star is a huge ball of hot gases that produces its own heat and light.

Q2. Why does the Sun look bigger than other stars?

Answer: The Sun looks bigger because it is much closer to the Earth than other stars.

Q3. What is a constellation?

Answer: A constellation is a group of stars that forms a recognisable pattern in the sky.

Q4. How many constellations are officially recognised?

Answer: The International Astronomical Union recognises 88 constellations.

Q5. What is a galaxy?

Answer: A galaxy is a huge group of stars, gas and dust held together by gravity.

Q6. Which is the nearest known star to the Sun?

Answer: Proxima Centauri is the nearest known star to the Sun.

Q7. What is a light-year?

Answer: A light-year is the distance travelled by light in one year. It is used to measure very large distances in space.

Q8. How were constellations useful in ancient times?

Answer: Travellers and sailors used stars and constellations to find directions and navigate their routes.

Q9. What is Ursa Major?

Answer: Ursa Major is a constellation of seven bright stars. It looks like a big spoon and is also called the Big Dipper or Great Bear.

Q10. What is Ursa Minor?

Answer: Ursa Minor is a constellation that contains the Pole Star. It is also called the Little Dipper or Lesser Bear.

Q11. What is special about the Pole Star?

Answer: The Pole Star appears nearly fixed in the northern sky and helps us find the north direction.

Q12. What is Orion?

Answer: Orion is a constellation represented as a hunter. Its three middle stars form the hunter's belt.

Q13. What is Cassiopeia?

Answer: Cassiopeia is a constellation of five bright stars arranged in a W or M shape.

Q14. What is Canis Major?

Answer: Canis Major is a constellation that contains Sirius, the brightest star in the night sky.

Q15. What is light pollution?

Answer: Light pollution is the excessive artificial light present at night. It makes stars difficult to see.

Q16. How can we reduce light pollution?

Answer: We can use only the necessary lights, switch off unused lights and direct outdoor lights towards the ground.

Q17. Why can we not see every constellation from every place?

Answer: The visibility of constellations depends on our location on Earth and the time of the year.

Q18. What is the Big Bang Theory?

Answer: The Big Bang Theory explains how the universe began and developed from an extremely hot and dense early state.

3. Telescope

Q1. What is a telescope?

Answer: A telescope is a device used to see distant objects clearly.

Q2. Why is a telescope important in astronomy?

Answer: A telescope helps scientists observe distant stars, planets, galaxies and other objects in space.

Q3. Who invented the telescope?

Answer: The chapter mentions that Galileo Galilei was an Italian astronomer who studied the universe with a telescope. He was not necessarily the inventor of the telescope.

Q4. Which telescope is mentioned as the biggest in the world in the chapter?

Answer: The chapter mentions the Keck Telescope in Hawaii.

Q5. What is astronomy?

Answer: Astronomy is the study of the universe and objects such as stars, planets, galaxies and asteroids.

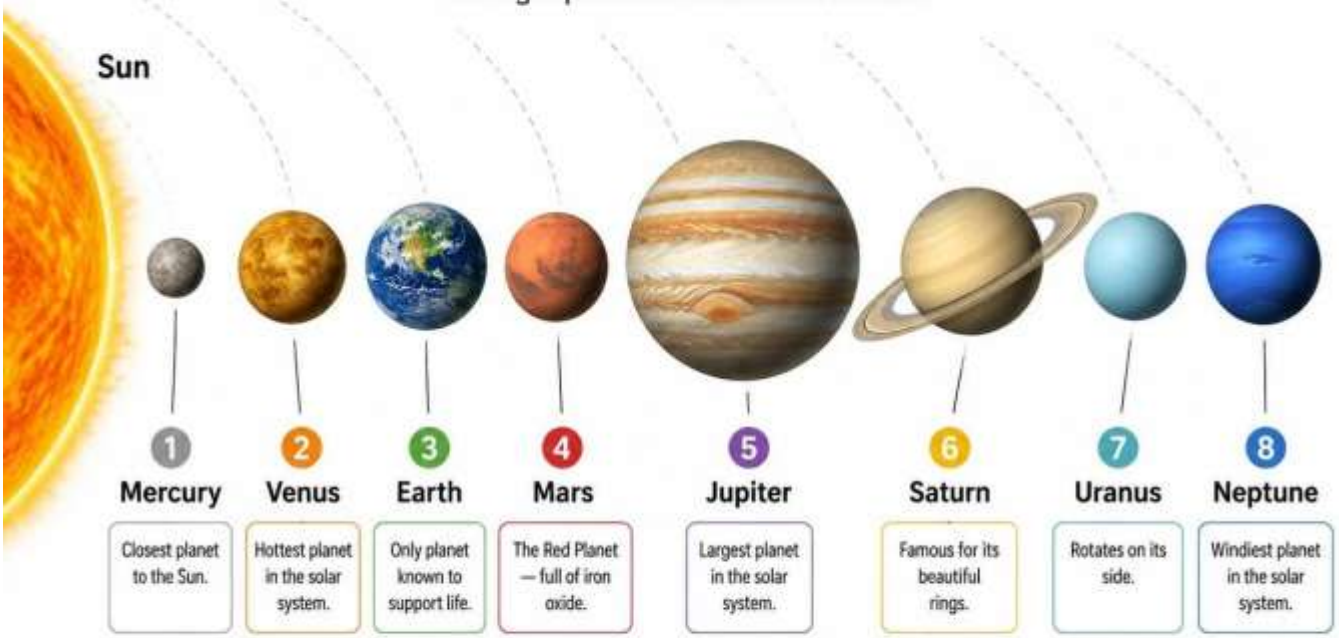
Q6. Who are astronomers?

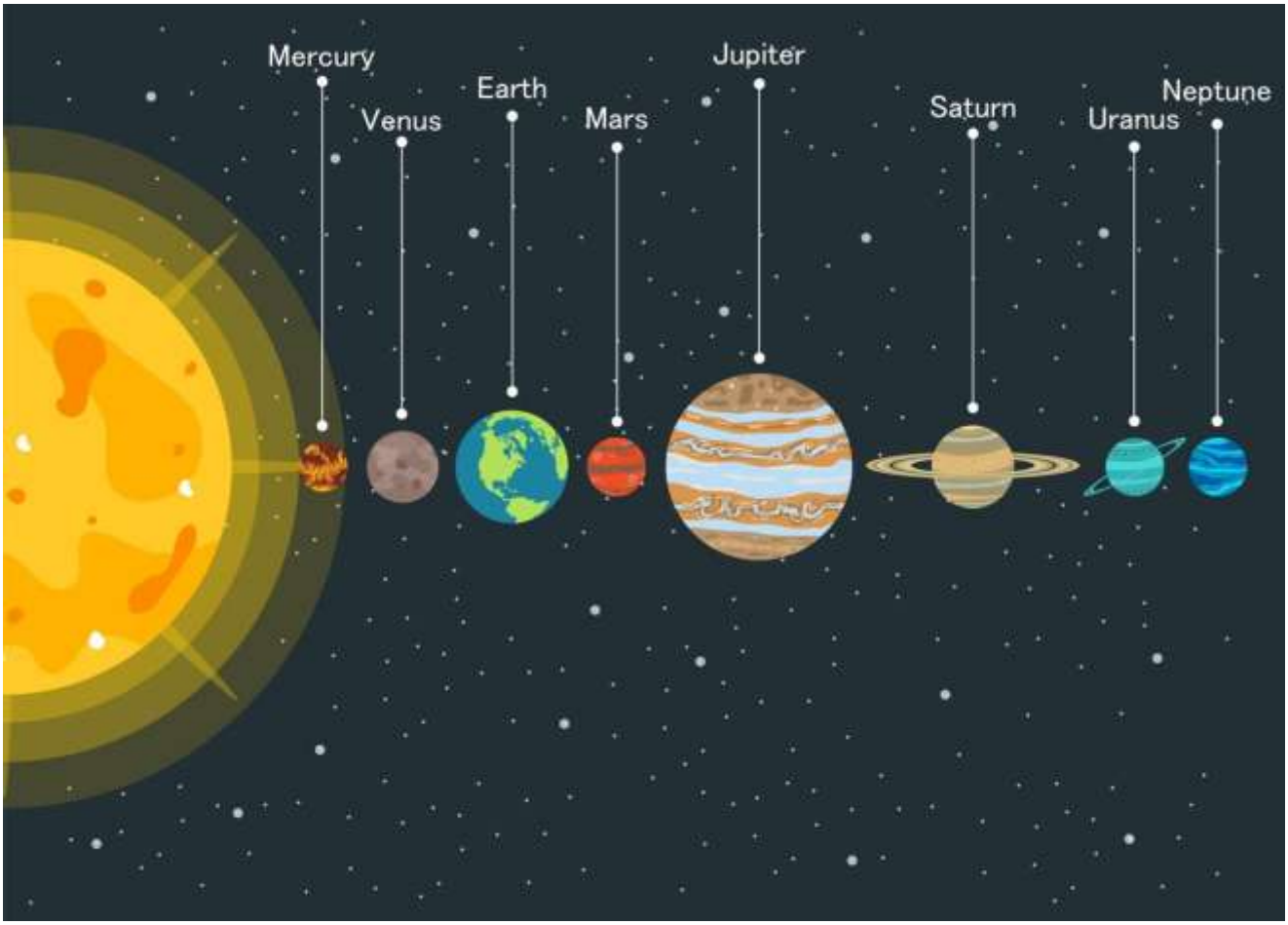
Answer: Astronomers are people who study objects and events in space.

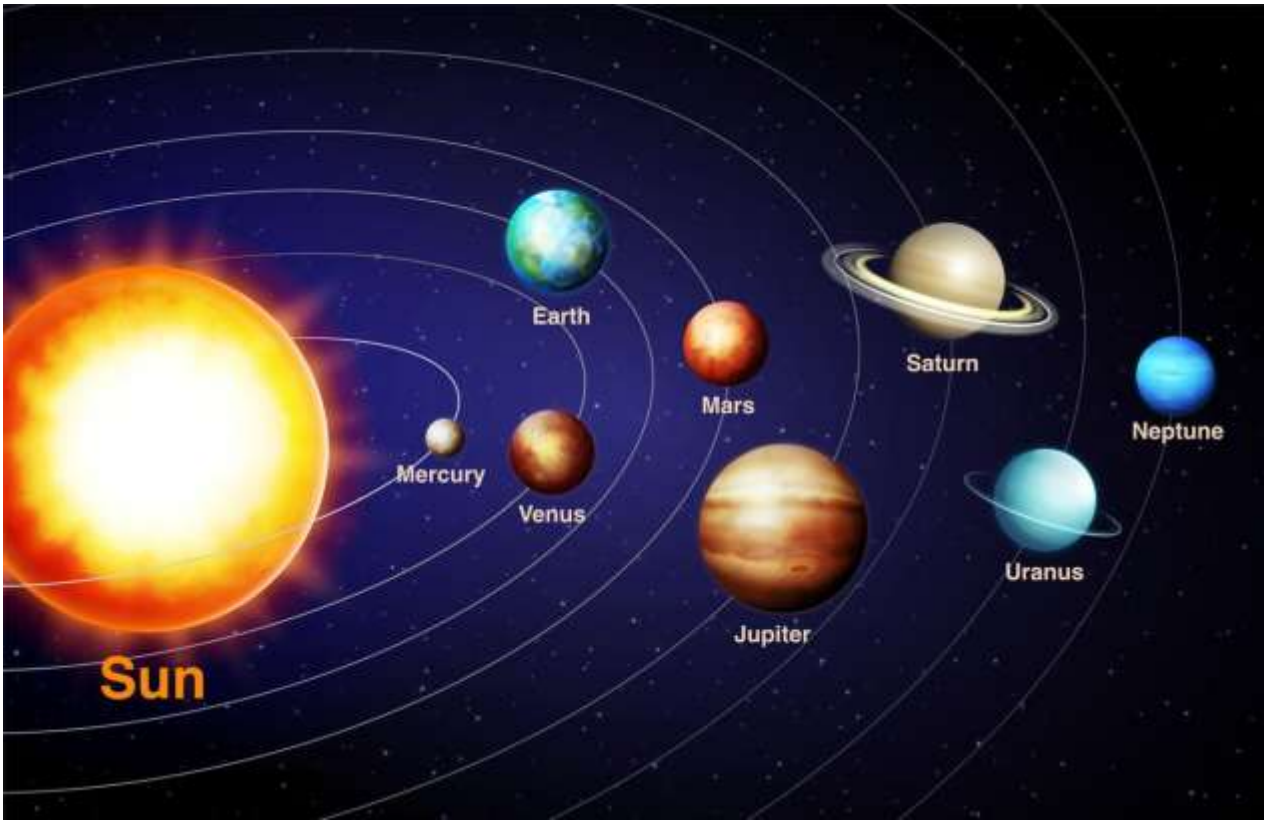
4. The Solar System

THE SOLAR SYSTEM

The eight planets in order from the Sun.




Mercury
Venus
Earth
Mars
Jupiter
Saturn
Uranus
Neptune



4

Q1. What is the solar system?

Answer: The solar system consists of the Sun, eight planets and other celestial bodies that revolve around the Sun.

Q2. How was the solar system formed according to the chapter?

Answer: The chapter explains that the solar system formed billions of years ago from a giant cloud of gas and dust.

Q3. Which body is at the centre of the solar system?

Answer: The Sun is at the centre of the solar system.

Q4. What keeps the planets moving around the Sun?

Answer: The Sun's gravitational pull keeps the planets in their orbits.

Q5. What is an orbit?

Answer: An orbit is the fixed path along which a planet or another celestial body moves around another body.

Q6. Name the eight planets in order from the Sun.

Answer: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus and Neptune.

Q7. What are the two main movements of planets?

Answer: The two main movements are rotation and revolution.

Q8. What is rotation?

Answer: Rotation is the spinning of an object on its own axis.

Q9. What is revolution?

Answer: Revolution is the movement of an object around another object.

5. The Sun

Q1. What is the Sun?

Answer: The Sun is a star and the largest object in our solar system.

Q2. Which gases mainly make up the Sun?

Answer: The Sun is mainly made up of hydrogen and helium.

Q3. How far is the Sun from the Earth?

Answer: The Sun is about 150 million kilometres away from the Earth.

Q4. How long does sunlight take to reach Earth?

Answer: Sunlight takes about 8 minutes to reach the Earth.

Q5. Why can we see the Sun?

Answer: We can see the Sun because it produces its own light.

Q6. Why is the Sun important for life on Earth?

Answer: The Sun provides light and heat. It supports plant growth and helps drive the water cycle and weather.

Q7. Why is the Sun called a star while the other stars are not visible during the day?

Answer: The Sun is a star because it produces its own light and heat. It appears much brighter than other stars because it is much closer to Earth. Its bright light makes other stars difficult to see during the day.

6. Planets

Q1. What is a planet?

Answer: A planet is a celestial body that revolves around the Sun in an orbit and does not produce its own light.

Q2. Why can we see planets?

Answer: We can see planets because they reflect the light of the Sun.

Q3. What are the three types of planets mentioned in the chapter?

Answer: They are inner planets, outer planets and dwarf planets.

Q4. Name the inner planets.

Answer: Mercury, Venus, Earth and Mars are the inner planets.

Q5. What are the main features of inner planets?

Answer: Inner planets are closer to the Sun. They are relatively small, rocky and dense.

Q6. Name the outer planets.

Answer: Jupiter, Saturn, Uranus and Neptune are the outer planets.

Q7. What are the main features of outer planets?

Answer: Outer planets are farther from the Sun and are generally larger. They are mostly made of gases and icy materials and some have prominent rings.

Q8. Name some dwarf planets.

Answer: Pluto, Ceres, Haumea, Makemake and Eris are dwarf planets mentioned in the chapter.

Q9. Why is Pluto called a dwarf planet?

Answer: Pluto is very small compared with the eight planets and is classified as a dwarf planet.

Q10. Which is the smallest planet?

Answer: Mercury is the smallest planet in the solar system.

Q11. Which is the largest planet?

Answer: Jupiter is the largest planet in the solar system.

Q12. Which planet is known as the Red Planet? Why?

Answer: Mars is known as the Red Planet because it appears reddish in colour.

Q13. Which planet is known as the Blue Planet? Why?

Answer: Earth is known as the Blue Planet because a large part of its surface is covered by water.

Q14. Which planet is called the Morning Star and the Evening Star?

Answer: Venus is called the Morning Star when it is visible before sunrise and the Evening Star when it is visible after sunset.

Q15. Which is the hottest planet, and why?

Answer: Venus is the hottest planet because its dense atmosphere traps heat.

Q16. Which planet has prominent rings?

Answer: Saturn is famous for its prominent rings made of dust, rock and ice.

Q17. Which planet is the farthest from the Sun?

Answer: Neptune is the farthest of the eight planets from the Sun.

Q18. Which planet is the second smallest?

Answer: Mars is the second smallest planet in the solar system.

Q19. Why is Neptune considered very cold?

Answer: Neptune is very far from the Sun and receives very little solar energy.

Q20. Why does Venus appear so bright in Earth's sky?

Answer: Venus reflects a large amount of sunlight and is relatively close to Earth, so it appears very bright.

Q21. Is life possible on planets other than Earth?

Answer: Scientists are still searching for evidence of life beyond Earth. We do not yet know whether life exists on other planets.

Important planet facts for revision

Planet	Important fact
Mercury	Smallest planet; closest to the Sun
Venus	Hottest planet; Morning Star and Evening Star
Earth	Blue Planet; supports known life
Mars	Red Planet
Jupiter	Largest planet
Saturn	Famous for its rings
Uranus	Its axis is tilted by about 90°
Neptune	Farthest planet from the Sun

7. Satellites

Q1. What is a satellite?

Answer: A satellite is a celestial body or object that revolves around a planet.

Q2. What are the two types of satellites?

Answer: The two types are natural satellites and artificial satellites.

Q3. What is a natural satellite?

Answer: A natural satellite is a naturally occurring celestial body that revolves around a planet.

Q4. Give two examples of natural satellites.

Answer: The Moon and Ganymede are examples of natural satellites.

Q5. What is an artificial satellite?

Answer: An artificial satellite is a machine made by humans and sent into space for a particular purpose.

Q6. What are the uses of artificial satellites?

Answer: Artificial satellites are used for communication, weather forecasting, navigation and collecting information about the Earth and space.

Q7. Name the first artificial satellite.

Answer: Sputnik 1 was the first artificial satellite. It was launched in 1957.

Q8. Name the first Indian artificial satellite.

Answer: Aryabhata was the first Indian artificial satellite. It was launched in 1975.

Q9. Name two Indian artificial satellites mentioned in the chapter.

Answer: Bhaskara I and Bhaskara II.

Q10. Which is the largest natural satellite in the solar system according to the chapter?

Answer: Ganymede, a natural satellite of Jupiter, is the largest natural satellite in the solar system.

8. The Moon







Q1. What is the Moon?

Answer: The Moon is the only natural satellite of the Earth.

Q2. How far is the Moon from the Earth?

Answer: The Moon is about 384,400 kilometres away from the Earth.

Q3. How does the Moon shine?

Answer: The Moon shines by reflecting sunlight. It does not produce its own light.

Q4. Does the Moon have an atmosphere?

Answer: No, the Moon does not have a substantial atmosphere like the Earth.

Q5. Is there liquid water on the Moon's surface according to the chapter?

Answer: The chapter states that there is no liquid water on the Moon's surface.

Q6. Why are the days and nights on the Moon extreme?

Answer: The Moon has no substantial atmosphere to retain and distribute heat. Therefore, its days are very hot and its nights are extremely cold.

Q7. What are craters?

Answer: Craters are hollow, bowl-shaped depressions on the surface of the Moon.

Q8. Can humans walk on the Moon?

Answer: Yes. Humans have landed and walked on the Moon.

Q9. Would it be easier to walk or jump on the Moon?

Answer: It would be easier to jump higher on the Moon because its gravity is much weaker than Earth's gravity.

Q10. What was Chandrayaan-1?

Answer: Chandrayaan-1 was India's first Moon mission. It was launched in 2008.

Q11. What is special about Chandrayaan-3?

Answer: Chandrayaan-3 made India the first country to achieve a landing near the Moon's south pole region in 2023.

Q12. Why do astronauts need special equipment on the Moon?

Answer: They need special equipment because the Moon lacks a breathable atmosphere and has extreme temperatures.

9. Other Celestial Bodies

A. Asteroids

Q1. What is an asteroid?

Answer: An asteroid is a rocky celestial body that revolves around the Sun.

Q2. Where are most asteroids found?

Answer: Most asteroids are found in the asteroid belt between Mars and Jupiter.

Q3. What is the asteroid belt?

Answer: The asteroid belt is a region between Mars and Jupiter that contains millions of rocky objects.

Q4. Which is the largest asteroid mentioned in the chapter?

Answer: Ceres is mentioned as the largest asteroid. It is classified as a dwarf planet.

Q5. What are planetoids?

Answer: Larger asteroids are sometimes called planetoids.

B. Comets

Q1. What is a comet?

Answer: A comet is an icy celestial body that revolves around the Sun.

Q2. Why does a comet develop a tail?

Answer: As a comet comes closer to the Sun, its ice heats up and releases gas and dust. This material forms a tail.

Q3. What are the two types of comets?

Answer: The two types are periodic comets and non-periodic comets.

Q4. What is a periodic comet?

Answer: A periodic comet returns at regular intervals as it revolves around the Sun.

Q5. What is a non-periodic comet?

Answer: A non-periodic comet is one that has been observed only once or does not return at regular, known intervals.

Q6. What is Halley's Comet?

Answer: Halley's Comet is a periodic comet that takes about 76 years to complete one revolution around the Sun.

Q7. When was Halley's Comet last seen, according to the chapter?

Answer: The chapter states that it was last seen in 1986.

C. Meteoroids, Meteors and Meteorites

Q1. What is a meteoroid?

Answer: A meteoroid is a small solid piece of debris from an asteroid or comet that moves through space.

Q2. What is a meteor?

Answer: A meteor is the bright streak of light produced when a meteoroid enters the Earth's atmosphere and heats up.

Q3. What is a meteorite?

Answer: A meteorite is a piece of a meteoroid that survives its journey through the atmosphere and reaches the Earth's surface.

Q4. Why do meteors look like shooting stars?

Answer: They produce bright streaks of light as they burn in the Earth's atmosphere.

Q5. What may happen when a meteorite hits the Earth?

Answer: It may create a hollow depression or crater where it falls.

Q6. Differentiate between a meteoroid, a meteor and a meteorite.

Term	Meaning
Meteoroid	A small rocky or metallic object moving through space
Meteor	A streak of light caused by an object entering the atmosphere
Meteorite	A fragment that reaches the Earth's surface

10. Milky Way Galaxy

Q1. What is a galaxy?

Answer: A galaxy is a large collection of stars, gas and dust held together by gravity.

Q2. What is the name of our galaxy?

Answer: Our galaxy is called the Milky Way Galaxy.

Q3. What is the Indian name for the Milky Way?

Answer: It is also called Akash Ganga.

Q4. Why is it called the Milky Way?

Answer: It appears in the sky as a faint, milky band of light.

Q5. What is the shape of the Milky Way Galaxy?

Answer: The Milky Way is a spiral-shaped galaxy.

Q6. Which is the closest major galaxy to the Milky Way?

Answer: The Andromeda Galaxy is the nearest major galaxy to the Milky Way.

Q7. What are the three types of galaxies mentioned in the chapter?

Answer: Spiral, elliptical and irregular galaxies.

11. The Universe

Q1. What is the universe?

Answer: The universe includes all space, time, matter, energy, stars, planets, galaxies and other celestial bodies.

Q2. Are there galaxies beyond the Milky Way?

Answer: Yes, there are many other galaxies beyond the Milky Way.

Q3. What are exoplanets?

Answer: Exoplanets are planets that orbit stars outside our solar system.

Q4. Has life been discovered on other planets?

Answer: The chapter says that scientists have not yet found confirmed evidence of life elsewhere in the universe.

Q5. Why do scientists study the universe?

Answer: Scientists study the universe to understand stars, planets, galaxies, how the universe developed and whether life exists elsewhere.

Q6. What is the Big Bang Theory?

Answer: The Big Bang Theory is the scientific explanation for the beginning and early development of the universe.

12. Textbook activities and “Think and Answer”

Q1. How might people in ancient times have used stars and constellations for navigation?

Answer: They observed the positions of stars and constellations to identify directions and find their routes while travelling on land or sea.

Q2. What methods can we use to reduce light pollution?

Answer: We can switch off unnecessary lights, use low-intensity lights and direct outdoor lights towards the ground.

Q3. Why is the Sun called a star? Why are other stars not visible during the day?

Answer: The Sun is a star because it produces its own heat and light. It is much closer to Earth than other stars. Its bright light makes the other stars difficult to see during the day.

Q4. Do you think life is possible on any other planet?

Answer: Life may be possible on other planets, but scientists have not yet confirmed the existence of life beyond Earth.

Q5. Why does Venus appear as the brightest planet in Earth's sky?

Answer: Venus reflects a large amount of sunlight and is relatively close to Earth, so it appears very bright.

Q6. Which is the nearest planet to Earth?

Answer: For the textbook's solar-system activity, the expected answer is Venus. However, the nearest planet varies with the planets' positions in their orbits; Mercury is the closest on average over time.

Q7. Which planet is the farthest from the Sun?

Answer: Neptune.

Q8. Which planet is called the Blue Planet?

Answer: Earth.

Q9. Which planet is called the Red Planet?

Answer: Mars.

13. Quick Check – Fill in the blanks

The following are the answers to the five questions on the last page of the PDF.

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1. Proxima Centauri is the nearest known star to the Sun.
2. Ursa Minor is also known as the Little Dipper.
3. Light pollution is the excessive amount of artificial light that appears at night.
4. The chapter gives the Keck Telescope as the world's biggest telescope.
5. The Moon is the only natural satellite of the Earth.

14. Additional important questions for exams

Q1. Differentiate between a star and a planet.

Star	Planet
A star produces its own heat and light.	A planet does not produce its own light.
A star is made of very hot gases.	Planets may be rocky or made mostly of gas and ice.
Example: Sun	Example: Earth

Q2. Differentiate between rotation and revolution.

Answer:

- Rotation: The spinning of a body on its own axis.
- Revolution: The movement of a body around another body.

Q3. Differentiate between natural and artificial satellites.

Answer: Natural satellites occur naturally, such as the Moon. Artificial satellites are made by humans and sent into space, such as Aryabhata.

Q4. Why is Earth suitable for life?

Answer: Earth has liquid water, a suitable atmosphere and temperatures that support life as we know it.

Q5. Why are outer planets generally colder than inner planets?

Answer: Outer planets are much farther from the Sun and receive less solar energy.

Q6. Why is the Moon's surface covered with craters?

Answer: Many craters were formed when space rocks struck the Moon's surface.

Q7. What is the difference between a comet and an asteroid?

Answer: A comet contains a large amount of ice, dust and rocky material and may develop a tail near the Sun. An asteroid is mainly rocky or metallic.

Q8. What is the difference between a solar system and a galaxy?

Answer: A solar system contains a star and the objects that revolve around it. A galaxy contains a huge number of stars and their planetary systems, along with gas and dust.

Q9. Why do we need telescopes?

Answer: Telescopes help us observe distant celestial objects that are difficult or impossible to see clearly with the naked eye.

Q10. What is the importance of artificial satellites in daily life?

Answer: They help us communicate, use navigation services, forecast weather and collect information about Earth.

15. Multiple-choice questions (MCQs)

Practice quiz

0 / 12 answered

1. Which is the nearest known star to the Sun?

A. Sirius
B. Proxima Centauri
C. Polaris
D. Betelgeuse

2. How many planets are there in the solar system?

A. Seven
B. Eight
C. Nine
D. Ten

3. Which is the largest planet?

A. Earth
B. Saturn
C. Jupiter
D. Neptune

4. Which planet is called the Red Planet?

A. Venus
B. Mars
C. Mercury
D. Uranus

5. Which is Earth's natural satellite?

A. The Sun
B. Mars
C. The Moon
D. Venus

6. Which constellation contains the Pole Star?

A. Ursa Minor
B. Orion
C. Canis Major
D. Cassiopeia

7. What is the name of our galaxy?

A. Andromeda
B. Milky Way
C. Orion
D. Ursa Major

8. Which planet is famous for its prominent rings?

A. Earth
B. Mercury
C. Saturn
D. Mars

9. What is a meteorite?

A. A star
B. A planet
C. A comet's tail
D. A space fragment that reaches Earth's surface

10. Which was the first Indian artificial satellite?

A. Sputnik 1
B. Aryabhata
C. Bhaskara II
D. Chandrayaan-1

11. What is a constellation?

A. A group of stars forming a recognisable pattern
B. A type of planet
C. A space machine
D. A moon crater

12. Which planet is the farthest from the Sun?

A. Jupiter
B. Uranus
C. Neptune
D. Saturn
Check answers

16. Fill in the blanks – Practice

1. The Sun is a _____.
2. The Earth revolves around the _____.
3. Jupiter is the _____ planet in the solar system.
4. The Moon reflects light from the _____.
5. Most asteroids are found between _____ and _____.
6. A comet may develop a _____ when it comes near the Sun.
7. The Milky Way is a _____-shaped galaxy.
8. Sputnik 1 was launched in the year _____.
9. The study of celestial objects is called _____.
10. A streak of light produced by a meteoroid in Earth's atmosphere is called a _____.

Answers: 1. Star 2. Sun 3. Largest 4. Sun 5. Mars, Jupiter 6. Tail 7. Spiral 8. 1957 9. Astronomy 10. Meteor

17. True or False

1. The Sun is a planet. — False
2. Earth is the third planet from the Sun. — True
3. All planets produce their own light. — False
4. Saturn is famous for its rings. — True
5. The Moon is Earth's natural satellite. — True
6. Halley's Comet is a periodic comet. — True
7. A meteorite always burns up completely before reaching Earth. — False
8. Proxima Centauri is the nearest known star to the Sun. — True
9. There are only eight galaxies in the universe. — False
10. Telescopes help us observe distant celestial objects. — True

18. One-line revision notes

- Star: A hot ball of gas that produces its own light.
- Constellation: A recognisable pattern of stars.
- Galaxy: A huge collection of stars, gas and dust.
- Telescope: A device used to observe distant objects.
- Solar system: The Sun and the celestial bodies that revolve around it.
- Orbit: The path followed by a celestial body around another body.
- Rotation: Spinning on an axis.
- Revolution: Movement around another body.
- Satellite: An object that revolves around a planet.
- Asteroid: A rocky celestial body that revolves around the Sun.
- Comet: An icy body that may form a tail near the Sun.
- Meteoroid: A small piece of debris moving through space.
- Meteor: A streak of light produced in the atmosphere.
- Meteorite: A space fragment that reaches Earth's surface.
- Exoplanet: A planet orbiting a star outside our solar system.

Exam tip: Learn the definitions, the eight planets in order, the differences between related terms, the names of important constellations and satellites, and the five Quick Check answers. These are especially useful for short-answer questions and MCQs.